

Discrete Structures

Proposition logic: either T or F $p \vee q, p \wedge q, \neg p, \neg q, p \oplus q$
 * Predicate logic: Represents in form of functions.
 $f(x) = x \oplus 3$
 Input $\rightarrow$ x $\rightarrow$ output $\rightarrow$ quantifier

Mathematical logic: is a tool for working with complicated compound statements. Includes: (1) Language for expressing $\wedge \vee$
 (2) concise notation for writing $p \rightarrow q$
 (3) Methodology for objectively reasoning about where truth or falsity T/F
 (4) It is the foundation for expressing formal proof in all branches of mathematics.

There are two types of logic $\begin{cases} \rightarrow \text{Proposition} \\ \rightarrow \text{Predicate} \end{cases}$

Proposition: * logic of compound statements built from simpler statements using so called boolean connectives.

Applications: (1) Design of digital electronics
 (2) Expressing conditi. in programs
 (3) queries to search engines & data bases

* A proposition (p, q, r) is simply a statement i.e a declarative sentence with a definite meaning, having a truth value that is either true or false

operator: binary, unary

operands: $2 \oplus 3$

Boolean operators: Work on propositions.

(1) Negate, $\neg$, NOT, UNARY

* p = you study hard
 q = you will get a good grade
 $p \rightarrow q$ = if you study hard you will get a good grade.

- (2) Conjunction, $\wedge$, AND, Binary
- (3) Disjunction, $\vee$, OR, Binary
- (4) Exclusive or, $\oplus$, XOR, Binary
- (5) Implication, $\rightarrow$, Implies, Binary
- (6) Biconditional, $\leftrightarrow$, If, Binary

* Negation

p	$\neg p$
T	F
F	T

x	y	$x \oplus y$ (XOR)
0	0	0
0	1	1
1	0	1
1	1	0

(AND)

x	y	$x \wedge y$
1	1	1
1	0	0
0	1	0
0	0	0

(Precedence)
 NOT
 AND
 OR

(OR)

x	y	$x \vee y$
1	1	1
1	0	1
0	1	1
0	0	0

p = It rained last night
 q = The sprinklers came on last night
 r = The lawn was wet this morning

- ① $\neg p$ = It did not rain last night
- ② $r \wedge \neg p$ = The lawn was ... and it did not ...
- ③ $\neg r \vee p \vee q$ = Either the lawn was not wet ... or ... or the ...
 when 2 $\vee$ then either or
 Either it ... or ... + but not both, - XOR (only one)
- $p \rightarrow q$: if p is true then q is true but if $p = \text{!T}$ then ...
 could be either T or F. - Implies *

x	y	F = $p \rightarrow q$
1	1	T
1	0	F
0	1	T
0	0	T

x	y	F = $p \leftrightarrow q = !x \vee !y$
1	1	1
1	0	0
0	1	0
0	0	1

$p \leftrightarrow q$: p & q have same truth value.

Some English phrases which implies $p \rightarrow q$

1. If p , then q
2. p implies q
3. If p , q
4. When p , q
5. Whenever p , q
6. q when p
7. q if p
8. p only if q
9. q is necessary for p
10. p is sufficient for q
11. q is implied by p
12. q follows from p

① converse: $q \rightarrow p$

② Inverse: $\neg p \rightarrow \neg q$

③ Contrapositive: $\neg q \rightarrow \neg p$

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
1	1	0	0	1	1	1	1
1	0	0	1	0	0	1	0
0	1	1	0	1	0	0	1
0	0	1	1	1	1	1	1

Propositional Equivalence: 2 syntactically diff. compound propositions may be syntactically identical i.e. have same meaning. called Equivalent

eg: $p \rightarrow q$; $\neg q \rightarrow \neg p$

Tautology: compound proposition i.e true no matter what the truth values of its atomic propositions are.

eg: $p \vee \neg p$

Contradiction: compound proposition which is always false.

Contingency: The other propositions except tautology & contradiction

Logical Equivalence: compound proposition p is logically equiv. to q if and only if $p \leftrightarrow q$ is a tautology. ($\Leftrightarrow$)

- If p and q are exactly same
- If p & q have same truth values.

Prove that $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$

p	q	$p \vee q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
1	1	1	0	1
1	0	1	0	1
0	1	1	0	1
0	0	0	1	0

Equivalence laws:-

1. Identity: $p \wedge T \Leftrightarrow p$, $p \vee F \Leftrightarrow p$
2. Domination: $p \vee T \Leftrightarrow T$, $p \wedge F \Leftrightarrow F$
3. Idempotent: $p \vee p \Leftrightarrow p$, $p \wedge p \Leftrightarrow p$
4. Double negation: $\neg\neg p \Leftrightarrow p$
5. Commutative: $p \vee q \Leftrightarrow q \vee p$, $p \wedge q \Leftrightarrow q \wedge p$
6. Associative: $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$
 $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$
7. Distributive: $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
 $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$
8. De Morgan's law: $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$
 $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$

$$p \rightarrow q \Leftrightarrow \neg p \vee q \quad \text{--- I}$$

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) \quad \text{--- II}$$

$$p \leftrightarrow q \Leftrightarrow \neg(p \oplus q) \quad \text{--- III}$$

same

same if different

$$p \wedge \neg q \rightarrow (p \oplus q)$$

$$\Leftrightarrow \neg(p \wedge \neg q) \vee (p \oplus q)$$

$$\Leftrightarrow \neg(p \wedge \neg q) \vee [(p \vee q) \wedge \neg(p \wedge q)]$$

$$Q. \quad p \wedge \neg q \rightarrow (p \oplus r) \Leftrightarrow \neg p \vee q \vee r$$

$$\neg(p \wedge \neg q) \vee (p \oplus r) \quad \text{--- using I}$$

$$(\neg p \vee q) \vee (p \oplus r)$$

p	q	$p \oplus r$
T	T	F
T	F	T
F	T	T
F	F	F

negation and

negation and

$$p \vee F = p$$

$$F \vee r = (r \vee F)$$

$$(p \vee r) \wedge \neg(p \wedge r)$$

$$(p \vee q \vee r)$$

$$(\neg p \vee q) \vee \{ (p \vee r) \wedge \neg(p \wedge r) \}$$

$$(\neg p \vee q \vee r)$$

$$\checkmark (\neg p \vee q) \vee \{ (p \vee r) \wedge (\neg p \vee \neg r) \}$$

$$(\neg p \vee q) \vee \{ (p \vee r \wedge \neg p) \vee (p \vee r \wedge \neg r) \}$$

proven

XXXXX

#

Ex: $P(x, y, z) \Rightarrow x$ gave y the grade z
 $x = \text{"Mike"}, y = \text{"Mary"}, \text{grade} = \text{"A"}$
 $x \in I$

Ex: $P(x) = "x+1 > x"$ $x \in I$
 "universe of discourse" - collection of values x can take in predicate

① Modus Ponens.

$\begin{array}{l} \text{if } (p \rightarrow q) \text{ is T} \\ \text{if } (p) \text{ is T} \\ \hline \text{then } (q) \end{array}$

② Modus Tollens

$\begin{array}{l} p \rightarrow q \\ \neg q \\ \hline \neg p \end{array}$

③ Disjunctive Syllogism

$\begin{array}{l} p \vee q \text{ is T} \\ \neg p \text{ is T} \\ \hline q \end{array}$

④ Hypothetical Syllogism

$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline p \rightarrow r \end{array}$

* only applicable to English statements

Predicate logic:

• generalises the grammatical notion of predicate to also include propositional functions of any no. of arguments, each of which may take any grammatical role.

Universe of discourse: the collection of values that a variable can take.

Quantifiers: provide a notation that allows us to quantify or count how many objects in the universe of discourse satisfy a given predicate.

① Universal Quantifier: $(\forall)$ for all

eg: $\forall x P(x)$

for all x this predicate is T.

② Existential Quantifier: ($\exists$) There exists
 $\exists x P(x)$

There exists at least one x for which the predicate is true

Variable:

Free $\leftarrow$

Bound $\leftarrow$

$\forall x P(x, y) = x$ gives money to y .

$\forall x \exists y P(x, y)$

Ex. $R(x, y) = x$ relies upon y domain: humans.

$\forall x (\exists y R(x, y)) =$ Everyone has someone to rely upon.

$\exists y (\forall x R(x, y)) =$ There is someone for everyone to rely upon.

$\exists x (\forall y R(x, y)) =$ There is someone who relies upon everyone.

$\forall y (\exists x R(x, y)) =$ Everyone is relied upon by someone.

$\forall x (\forall y R(x, y)) =$ Everyone relies upon Everyone.

$$(\neg p \vee q) \vee \{ (p \vee r) \wedge (\neg p \vee \neg r) \}$$

$$(\neg p \vee q) \vee \{ ((p \vee r) \wedge (\neg p)) \vee ((p \vee r) \wedge (\neg r)) \}$$

$$(\neg p \vee q) \vee \{ [(\neg p \wedge p) \vee (\neg p \wedge \neg r)] \vee [(\neg p \wedge r) \vee (\neg r \wedge r)] \}$$

$$(\neg p \vee q) \vee \{ (\neg p \wedge \neg r) \vee (\neg r \wedge p) \}$$

$$\begin{aligned} & [(\neg p \wedge \neg r) \vee (\neg r \wedge p)] \wedge [(\neg p \wedge r) \vee (\neg r \wedge p)] \\ & [(\neg r \vee \neg p) \wedge (\neg r \vee p)] \wedge [(\neg p \vee r) \wedge (p \vee r)] \\ & [(\neg r \vee \neg p) \wedge (p \vee r)] \end{aligned}$$

$$\begin{aligned}
 & \neg(p \wedge \neg q) \vee ((p \vee \neg p) \wedge \neg(p \wedge \neg q)) \\
 & (\neg p \vee q) \vee ((p \vee \neg p) \wedge \neg(p \wedge \neg q)) \\
 & (q \vee \neg p) \vee ((p \vee \neg p) \wedge \neg(p \wedge \neg q)) \\
 & q \vee (\neg p \vee ((p \vee \neg p) \wedge (\neg p \vee \neg q))) \\
 & q \vee (\overbrace{(\neg p \vee p \vee \neg q)}^T) \wedge (\neg p \vee \neg q) \\
 & q \vee (\underbrace{(\neg p \vee \neg q)}_{\text{Idempotent}}) \\
 & q \vee (\neg p \vee \neg q)
 \end{aligned}$$

R: Everybody likes somebody.

$$\forall x (\exists y R(x, y))$$

$$\# \forall x (P(x) \wedge Q(x))$$

$$\# \forall x (P(x) \wedge Q(x)) \quad // \text{Bound}$$

$$\# \forall x P(x) \wedge \exists x Q(x) \quad \checkmark$$

$$\# \forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y) \quad (\text{if same } \forall / \exists \checkmark)$$

$$\# \forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

$$\# \exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$$

H(x) = x is human

M(x) = x is mortal

G(x) = x is god

Premises

$$\forall x H(x) \rightarrow M(x)$$

$$\forall x G(x) \rightarrow \neg M(x)$$

$$\neg \exists x (H(x) \wedge G(x))$$

// humans are mortal

// gods are immortal

// There exists no human who is god.

Translate the following into English:

$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$ Domain: real no.
 $\Rightarrow$ The product of a positive real no and a negative real no is always negative.

$\Rightarrow$ The sum of 2 +ve integers is always +ve.

$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x+y > 0))$ Domain: Integers.
 $\forall x \forall y (x+y > 0)$ Domain: positive integers

$\Rightarrow$ For every real no. x if $x \neq 0$ then there is a real no. y such that $xy = 1$

$\forall x \forall y ((x \neq 0) \rightarrow \exists y (xy = 1))$

$\Rightarrow \forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$

$C(x)$: x has a computer

$F(x, y)$: x & y are friends.

Everyone has a comp. or a friend with a comp.

$\Rightarrow$ If a person is a student & is C.S major, then this person takes a course in mathematics.

$S(x)$: x is a student

$C(x)$: x is a C.S major

$P(x, y)$: x takes course in y

x : all people

y : ~~some~~ courses in math.

$\exists x \forall y ((S(x) \wedge C(x)) \rightarrow \exists y P(x, y))$

Everyone has exactly one best friend

$B(x, y)$: y is BFF of x .

$\forall x (\exists y (y = 1) \rightarrow B(x, y))$

$\exists x \exists y (B(x, y) \wedge \forall z (y \neq z \rightarrow \neg B(x, z)))$

$\rightarrow$ disjunctive

$P(x) \vee Q(x)$

$\rightarrow$ conjunctive

$P(x) \wedge Q(x)$

Disjunctive
 $\wedge$ disjunctive

Obtain the

$(p \vee q)$

$\neg(p \wedge q)$

Obtain the

Conjunctive

$\wedge$ statement

disjunctive

→ Disjunctive normal form:

$$P(x) \vee Q(x) \vee R(x)$$

^ can be combination $(p \vee q \wedge r)$

→ Conjunctive normal form:

$$P(x) \wedge Q(x) \wedge R(x)$$

Disjunctive Normal form:-

A disjunction of fundamental conjunctions

$$(p \wedge q \wedge r) \vee (q \wedge r) \vee (p \wedge q)$$

$$(p \wedge q) \vee \neg q$$

$$(\neg p \wedge q) \vee (q \wedge r)$$

Obtain the DNF for $(p \rightarrow q) \wedge (\neg p \wedge r)$

$$(\neg p \vee q) \wedge (\neg p \wedge r)$$

$$\neg p \wedge (\neg p \wedge r) \vee q \wedge (\neg p \wedge r)$$

$$(\neg p \wedge r) \vee (q \wedge \neg p)$$

Obtain the DNF for $\neg(p \rightarrow (q \wedge r))$

$$\neg(\neg p) \vee (q \wedge r)$$

$$\neg(\neg p \vee (q \wedge r))$$

$$\neg p \wedge \neg(q \wedge r)$$

$$p \wedge \neg(q \wedge r)$$

$$p \wedge (\neg q \vee \neg r)$$

$$(p \wedge \neg q) \vee (p \wedge \neg r)$$

Conjunctive Normal form:-

A statement which consists of conjunction of fundamental disjunctions.

$$p \wedge r; \neg p \wedge (p \vee q)$$

Obtain the CNF:

$$(p \wedge q) \vee (\neg p \wedge q \wedge r)$$

$$p \vee (\neg p \wedge q \wedge r) \wedge (q \vee (\neg p \wedge q \wedge r))$$

$$((p \vee \neg p) \wedge (p \vee q) \wedge (p \vee r)) \wedge ((q \vee \neg p) \wedge (q \vee q) \wedge (q \vee r))$$

$$\cancel{(p \vee q) \wedge (p \vee r)} \wedge ((q \vee \neg p) \wedge (q \wedge q) \wedge (q \wedge r))$$

$$\cancel{(p \wedge q \wedge r)} \wedge ($$

$$((p \vee q) \wedge (p \vee r)) \wedge ((q \vee \neg p) \wedge (q \vee r) \wedge q)$$

$$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$$

$$(p \vee r) \wedge ((p \rightarrow q) \wedge (q \rightarrow p))$$

$$(p \vee r) \wedge ((\neg p \vee q) \wedge (\neg q \vee p))$$

$$(p \vee r) \wedge (\neg p \vee q) \wedge (\neg q \vee p)$$

Find the DNF for:

p	q	r	$f(p, q, r)$	
F	F	F	T	✓
F	F	T	F	
F	T	F	T	✓
F	T	T	F	
T	F	F	T	✓
T	F	T	F	
T	T	F	F	
T	T	T	T	✓

$$(\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r) \vee (p \wedge q \wedge r)$$

 p : food is good.
 q : service is good
 r : Restaurant is 5*

1. Write the following in symbolic notations:

* Either food is good or service is good or both

* Either food is good or service is good not both

* food is good while service is poor

* If both the food & service are good then the rating will be 5*.

* It is not true that a 5* rating always mean good food and good service.

1. $(p \vee q) \vee (p \wedge q)$

2. $p \oplus q : (p \vee q) \wedge \neg (p \wedge q)$

3. $p \wedge \neg q$

4. $\neg (p \wedge r)$

5. $(p \wedge q) \rightarrow r$

6. $\neg (r) \rightarrow (p \wedge q)$

Find the truth value of:

$$[p \rightarrow ((q \wedge (\neg r)) \vee s)] \wedge [(\neg x) \leftrightarrow (s \wedge r)]$$

where x is false & p, q, r, s are true

$A =$

$B =$

$A = p \rightarrow A' \quad A = T \rightarrow T = T$

$A' = (q \wedge (\neg r)) \vee s$

$(T \wedge F) \vee T$

$F \vee T$

$\rightarrow T$

$B = (\neg x) \leftrightarrow (s \wedge r)$

$= T \leftrightarrow T$

$= T$

* Show that $(p \wedge (p \rightarrow q)) \rightarrow q$ is tautology.

$$(p \wedge (\neg p \vee q)) \rightarrow q$$

$$\neg (p \wedge (\neg p \vee q)) \vee q$$

$$\neg ((p \wedge \neg p) \vee (p \wedge q)) \vee q$$

$$\neg (p \wedge q) \vee q$$

$$\neg p \vee \neg q \vee q$$

I

* CNF, DNF for $(\neg p \vee q) \leftrightarrow (p \wedge q)$

p	q	$\neg p \vee q$	$p \wedge q$	f
1	1	1	1	1
1	0	0	0	1
0	1	1	0	0
0	0	1	0	0

$$\rightarrow (p \wedge q) \vee (\neg p \wedge \neg q) - \text{DNF}$$

$$\rightarrow (\neg p \vee q) \wedge (\neg p \vee \neg q) - \text{CNF}$$

$$\bullet (\neg p \vee q) \rightarrow (p \wedge q) \wedge (p \wedge q) \rightarrow (\neg p \vee q)$$

* If I am not in good mood or I am not busy then I will go for movies.

$$(\neg p \vee \neg q) \rightarrow r$$

$$(\neg (p \wedge q)) \rightarrow r$$

* If you know C++ & Oracle then you will get a job

$$(p \wedge q) \rightarrow r$$

* I will score good marks iff I study hard

$$p \leftrightarrow q$$

* program is readable only if it is well structured.

$$p \leftrightarrow q$$

* Unless he studies he will fail in examination.

p: he studies

q: he ~~pass~~ passes

$$\neg p \rightarrow \neg q$$

* There will be no test exam tomorrow if prof. is out of town or transportation is out of service.

$\neg p$: test tom.

q: prof. out

r: transportat. is working

$$(q \vee \neg r) \rightarrow \neg p$$